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Instructions

1.

You are developing a convolutional neural network (CNN) for classifying audio signals and notice that the model performs well on training data but poorly on new, unseen data. What is the primary reason for evaluating the performance of the CNN?

1 / 1 point

☐ To improve the quality of the audio input signals

☐ To reduce training time by optimizing the model architecture

☐ To determine the maximum number of layers that can be added

☒ To ensure the model generalizes well to unseen data and is not overfitting

✔ Correct

Correct! Evaluating performance helps identify if the model generalizes well to unseen data, which is crucial for practical applications and prevents overfitting.
2.

In the preprocessing phase for generating spectrograms, you decide to standardize the length of all audio signals. How does this affect the subsequent classification using a CNN?

1 / 1 point

☒ It ensures uniform feature representation across all audio samples, which improves the consistency of the input to the network.

☐ It increases the sensitivity of the spectrogram to high-frequency components.

☐ It reduces the computational complexity of the model during training.

☐ It enhances the signal-to-noise ratio of the audio signal.

✔ Correct

Correct! Standardizing the length of audio signals ensures that each sample has a uniform feature representation, which is crucial for effective classification.
3.

When designing a CNN for audio signal classification, you need to select various hyperparameters. Which of the following hyperparameters are crucial for optimizing the performance of your model? (Select all that apply)

1 / 1 point

☐ Using a fixed number of layers regardless of performance

☐ Setting the input data format to grayscale for consistency

☒ Implementing dropout to reduce overfitting and improve generalization

✔ Correct

Correct! Dropout is an essential technique to prevent overfitting and enhance the model's ability to generalize.

☒ Selecting the activation function to introduce non-linearity and improve learning

✔ Correct

Correct! The choice of activation function affects the network's ability to learn complex patterns.

☒ Choosing the kernel size to effectively capture relevant features

✔ Correct

Correct! The kernel size is crucial as it determines the receptive field and the scale of features captured by the convolutional layers.
4.

In the process of generating spectrograms, you apply several preprocessing steps. How do these steps collectively impact the quality and effectiveness of the CNN in audio classification tasks?

1 / 1 point

☐ Normalizing the length of audio signals to introduce variability and challenge the model

☒ Applying the Short-Time Fourier Transform (STFT) for time-frequency representation, normalizing amplitude to handle varying signal strengths, and padding audio signals to a fixed length all contribute to uniform and effective input features for the CNN

☐ Using Principal Component Analysis (PCA) to reduce dimensionality and improve feature extraction

☐ Applying a high-pass filter to enhance the high-frequency components and improve model performance

✔ Correct

Correct! These steps ensure that the input features are consistent and suitable for CNN processing, enhancing the model's ability to classify audio accurately.
5.

In a CNN designed for audio signal classification, you observe that adding more convolutional layers has led to diminishing returns on model performance. What could be a potential reason for this observation?

1 / 1 point

☒ The model may be overfitting due to increased complexity from additional layers, leading to poorer generalization on new data.

☐ The pooling layers might be too shallow to capture sufficient features.

☐ The activation functions are not sufficiently diverse across layers.

☐ The dropout rate might be too high, leading to underfitting.

✔ Correct

Correct! Adding too many layers can result in overfitting, where the model becomes too complex and performs poorly on new, unseen data.